Homework 1

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MA 542 Section A1

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Problem 1

$$12 \cdot 16 = 192 = 24 \cdot 8 \implies 12 \cdot 16 \equiv 0 \pmod{24}.$$

Problem 3

$$11 \cdot (-4) = -44 = (-3 \cdot 15) + 1 \implies 11 \cdot (-4) \equiv 1 \pmod{15}.$$

Problem 5

$$(2,3)(3,5) = (6,15) \equiv (1,6) \pmod{5 \times 9}.$$

Problem 7

The ring is commutative because it is a subring of $\mathbb{Z}$, and it has unity if and only if $n \in \{0,1\}$.

Problem 9

It's commutative because $\mathbb{Z}$ is commutative, and it has $(1,1)$ as a unity.

Problem 11

The ring is commutative, and has 1 as a unity. However, it is not a field. Let $a + b\sqrt{2}$ be the multiplicative inverse of 2; that is,

$$2\left(a + b\sqrt{2}\right) = 1.$$

Suppose for contradiction that $a + b\sqrt{2} \in \mathbb{Z}\left[\sqrt{2}\right]$. It follows that

$$2a + 2b\sqrt{2} = 1 \implies 2b\sqrt{2} = 1 - 2a.$$

Since $1 - 2a \in \mathbb{Z}$, we have that $2b\sqrt{2} \in \mathbb{Z}$. However, since $2b \in \mathbb{Z}$, we have that $2b\sqrt{2} \in \mathbb{Z}$ if and only if $b = 0$. It follows that

$$1 - 2a = 0 \implies a = \frac{1}{2} \notin \mathbb{Z} \implies \Longleftarrow\ .$$

Problem 13

Problem 15

Anything of the form $(0, n)$ or $(n, 0)$ is a zero divisor and thus not a unit. Additionally, the inverse of (a, b) if $a, b \neq 0$ is $(1/a, 1/b)$, and $1/a, 1/b \in \mathbb{Z}$ if and only if $a, b = 1$.

$$\{(1, 1)\} .$$

Problem 17

$\forall r \in \mathbb{Q} \setminus \{0\}$, $\exists 1/r \in \mathbb{Q} \setminus \{0\}$.

$$\mathbb{Q} \setminus \{0\} .$$

Problem 19

$a \in \mathbb{Z}_4$ is a unit if and only if a is nonzero and does not divide 4. It follows that $U(\mathbb{Z}_4) = \{1, 3\}$.

Problem 27

The conclusion that

$$(X - I_3)(X + I_3) = 0 \implies X \in \{-I_3, I_3\}$$

does not necessarily follow in $M_3(\mathbb{R})$, because $M_3(\mathbb{R})$ has zero divisors, so the cancellation law does not necessarily hold. For example, consider the matrix

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} .$$

Note that

$$X^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^2 = I_3.$$

However, $X \neq \pm I_3$.

Problem 39

Let $a, b, c \in R$. It follows that

$$a(bc) = a0 = 0, \qquad (ab)c = 0c = 0.$$

Therefore, associativity holds. Since $0 \in R$, closure holds. Finally, note that

$$a(b + c) = 0, \qquad ab + ac = 0 + 0 = 0,$$

$$(a + b)c = 0, \qquad ac + bc = 0 + 0 = 0.$$

Therefore, distributivity holds, and $(R, +, \cdot)$ is a ring.

Problem 41

Using the notation as in the problem, and assuming for now that $a, b \neq 0$, note that since p is prime, a, b are units, and thus $a, b \in U(\mathbb{Z}_p)$. Since p is prime, the order of $U(\mathbb{Z}_p)$ when viewed as a group is $p-1$. It follows that

$$a^{p-1} = b^{p-1} = 1.$$

Therefore, $a^p = a$ and $b^p = b$. Suppose $b \neq -a$. It follows that $(a+b)$ is also an element of $U(\mathbb{Z}_p)$, so we have $(a+b)^p = a+b$ by the same logic. Therefore, $(a+b)^p = a^p + b^p$.

If $b = -a$, then we have $(a-a)^p = 0$. If p is odd, then $(-a)^p = -a^p$, so we have $a^p - a^p = 0$. If $p = 2$, then $a = 1$, so $1^2 + 1^2 = 0 = (1-1)^2 \pmod 2$.

Finally note that if a or $b = 0$, then we have $(a+0)^p = a^p + 0^p$ or $(0+b)^p = 0^p + b^p$ trivially.

Problem 49